Spatial scRNA-seq

Jaewoong Lee Semin Lee

Department of Life Sciences
Gwangju Institute of Science and Technology

jaewoong@gist.ac.kr

2026-10-15

Overview

1. Abstract
2. Introduction
3. Materials
4. Methods
5. Results
6. References

1. Abstract

2. Introduction

3. Materials

4. Methods

Methods I

Basic method

1. Scanpy (Wolf, Angerer, & Theis, 2018)

Projection method

1. UMAP (McInnes, Healy, & Melville, 2018; Becht et al., 2019)
2. tSNE (Maaten & Hinton, 2008)
3. PHATE (Moon et al., 2019)
4. trimap (Amid & Warmuth, 2019)

Clustering

1. Leiden (Traag, Waltman, & van Eck, 2019)
2. Louvain (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008)

5. Results

5. Results

5.1. Quality check

Total counts per cell I

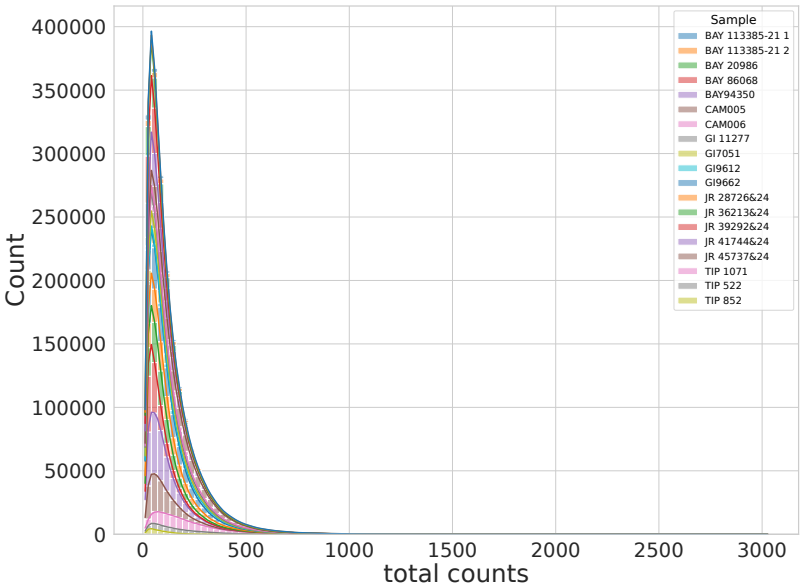


Figure: Total counts

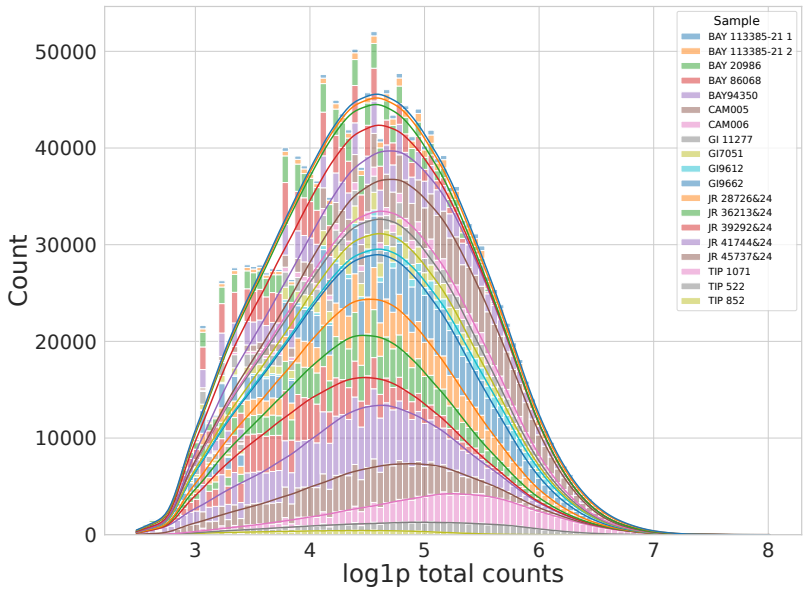


Figure: log1p(Total count)

Total counts per cell II

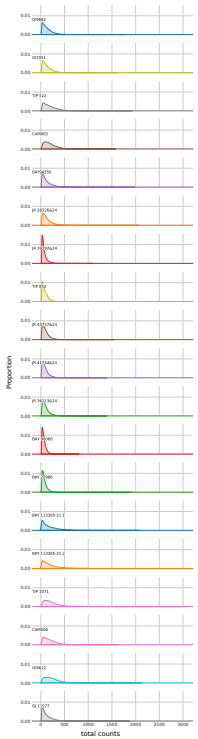


Figure: Genes by counts

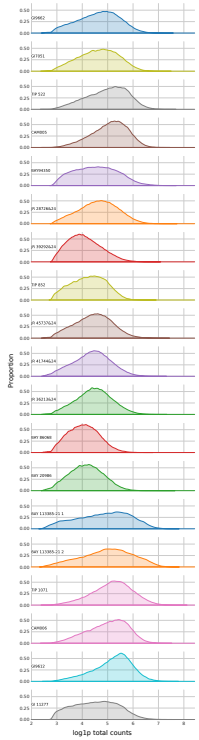


Figure: log1p(Genes by count)

Total counts per cell III

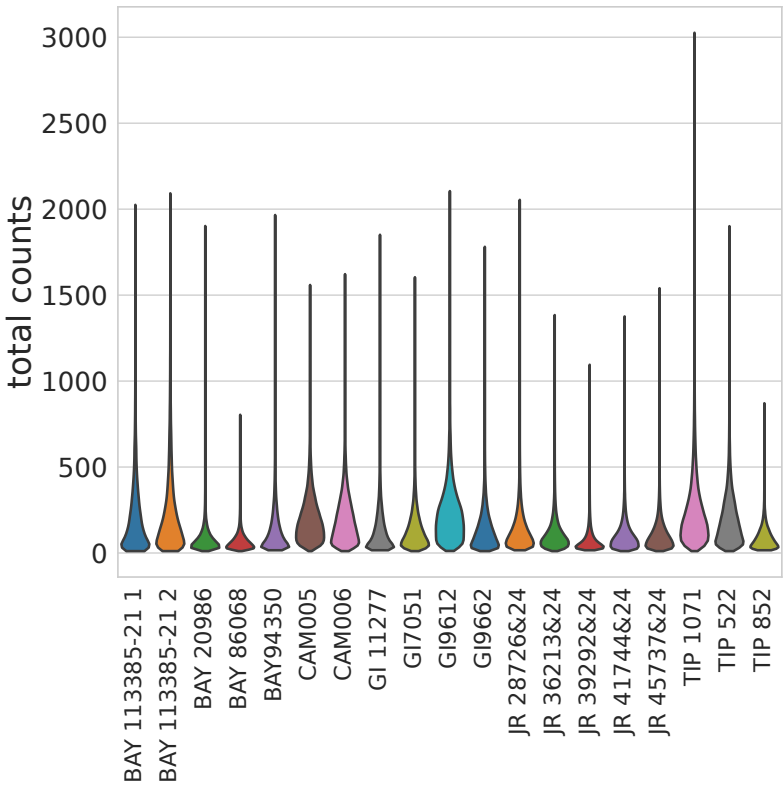


Figure: Genes by counts

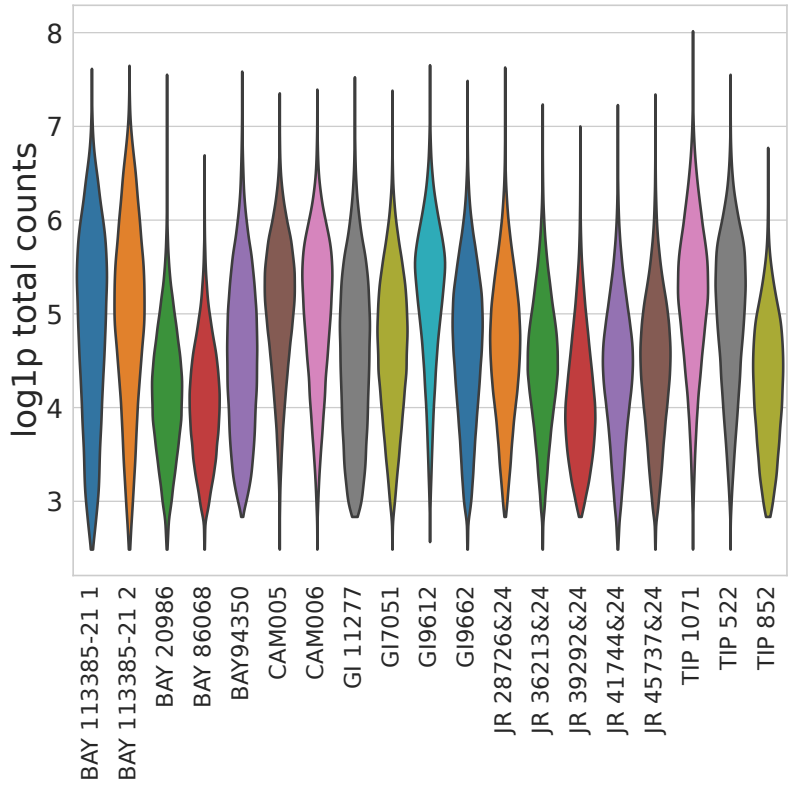


Figure: log1p(Genes by count)

Genes by counts I

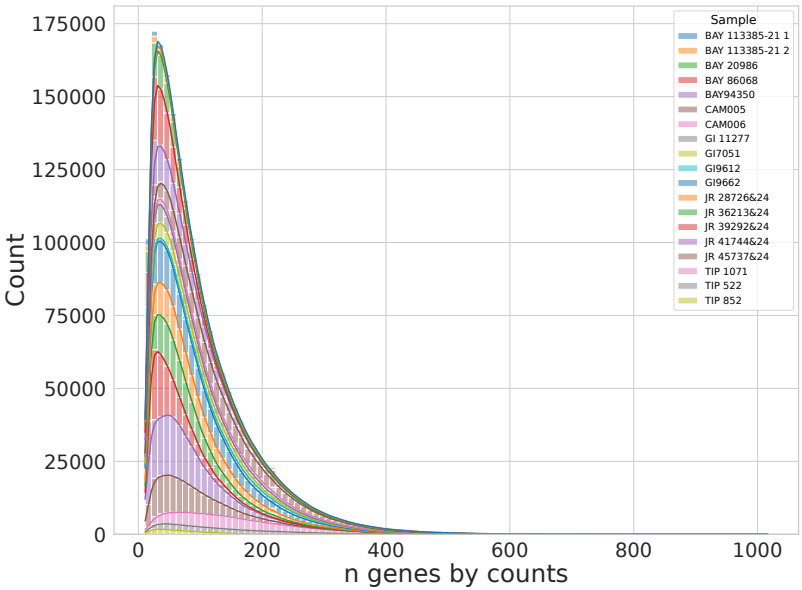


Figure: Genes by counts

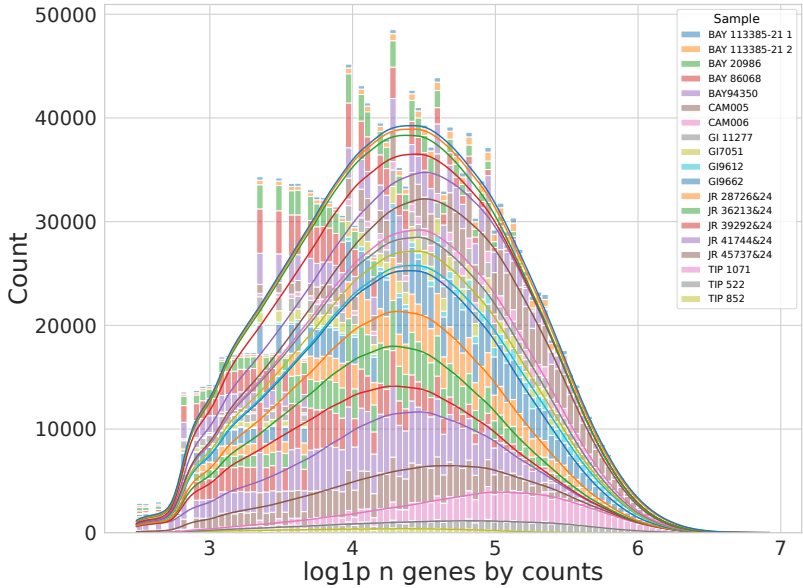


Figure: log1p(Genes by count)

Genes by counts II

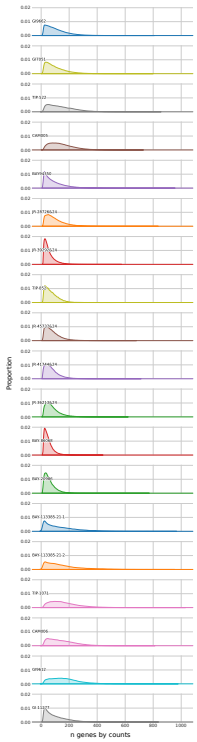


Figure: Genes by counts

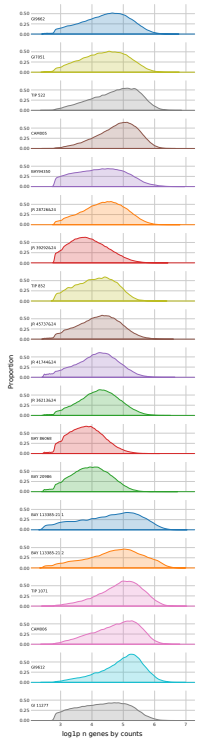


Figure: log1p(Genes by count)

Genes by counts III

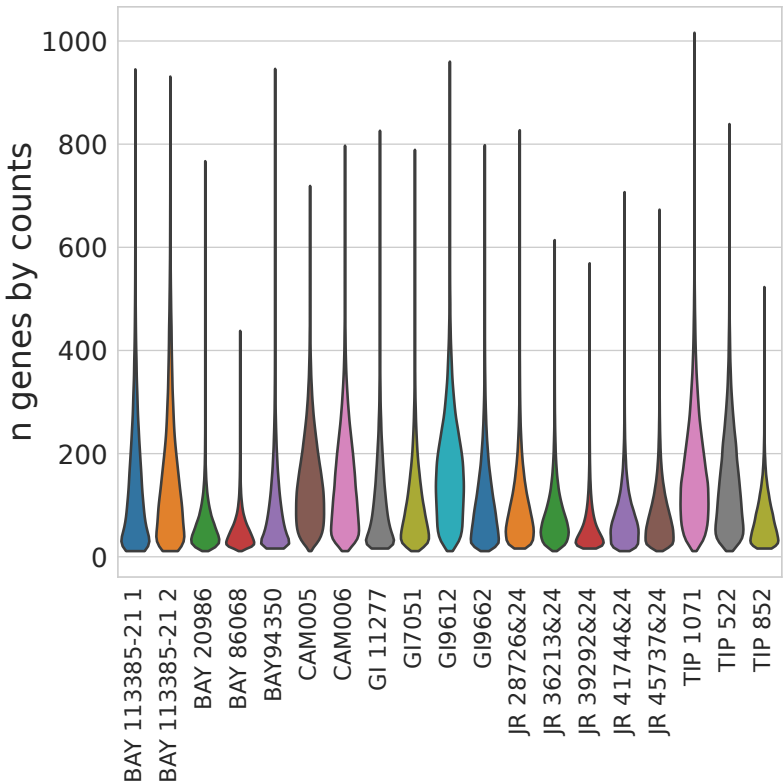


Figure: Genes by counts

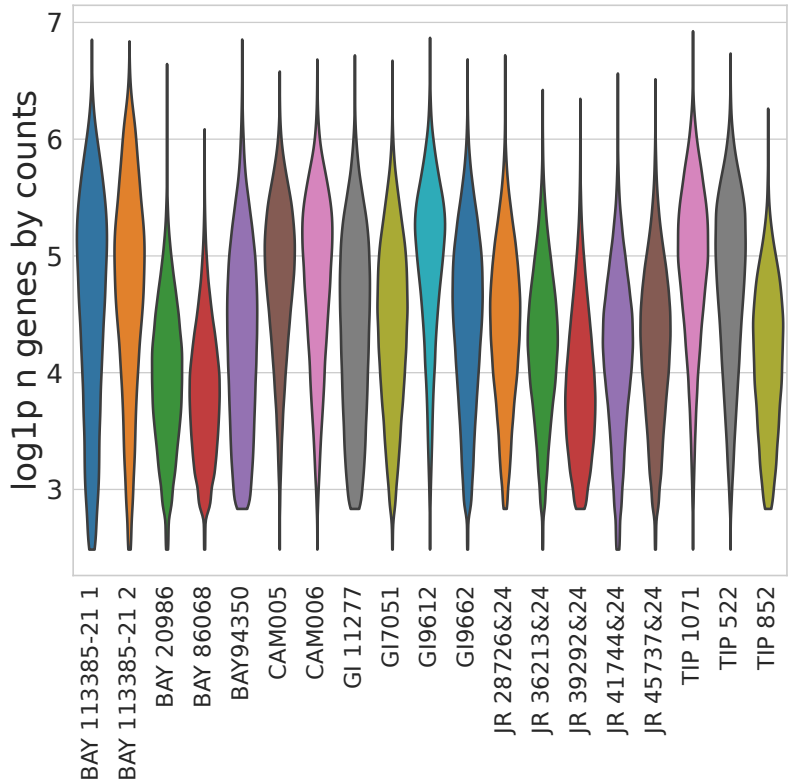


Figure: log1p(Genes by count)

5. Results

5.2. Highly variable genes

5. Results

5.2. Highly variable genes

5.2.1. Normalized dispersions

Highly variable genes with normalized dispersions I

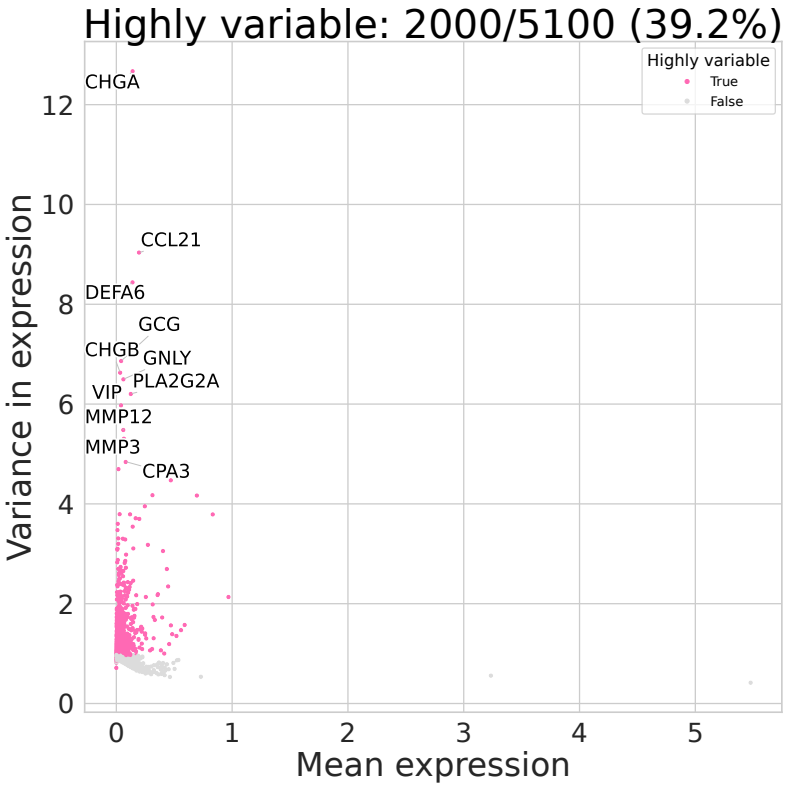


Figure: Volcano plot

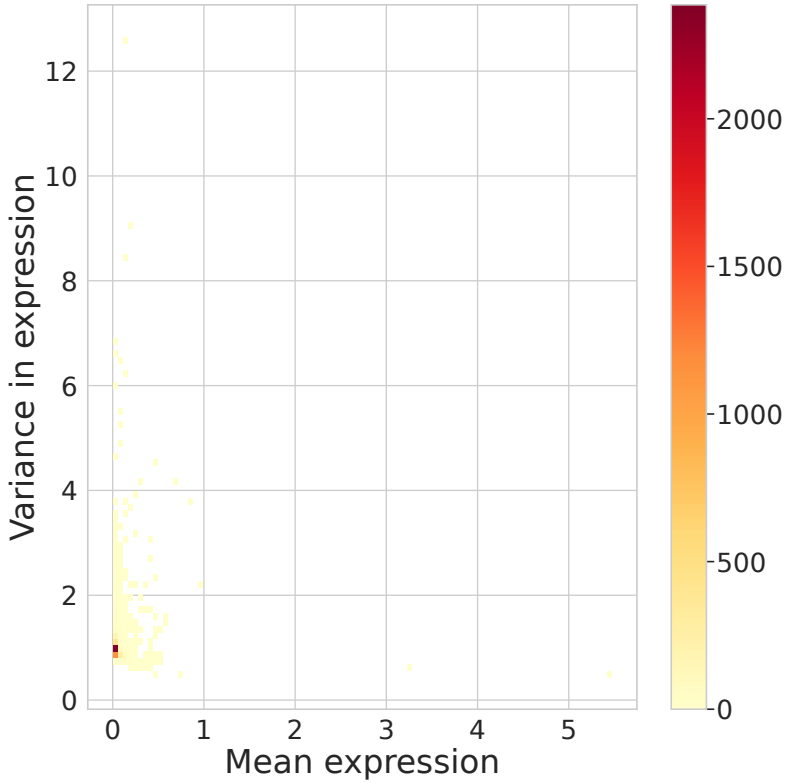


Figure: Distribution

Highly variable genes with normalized dispersions II

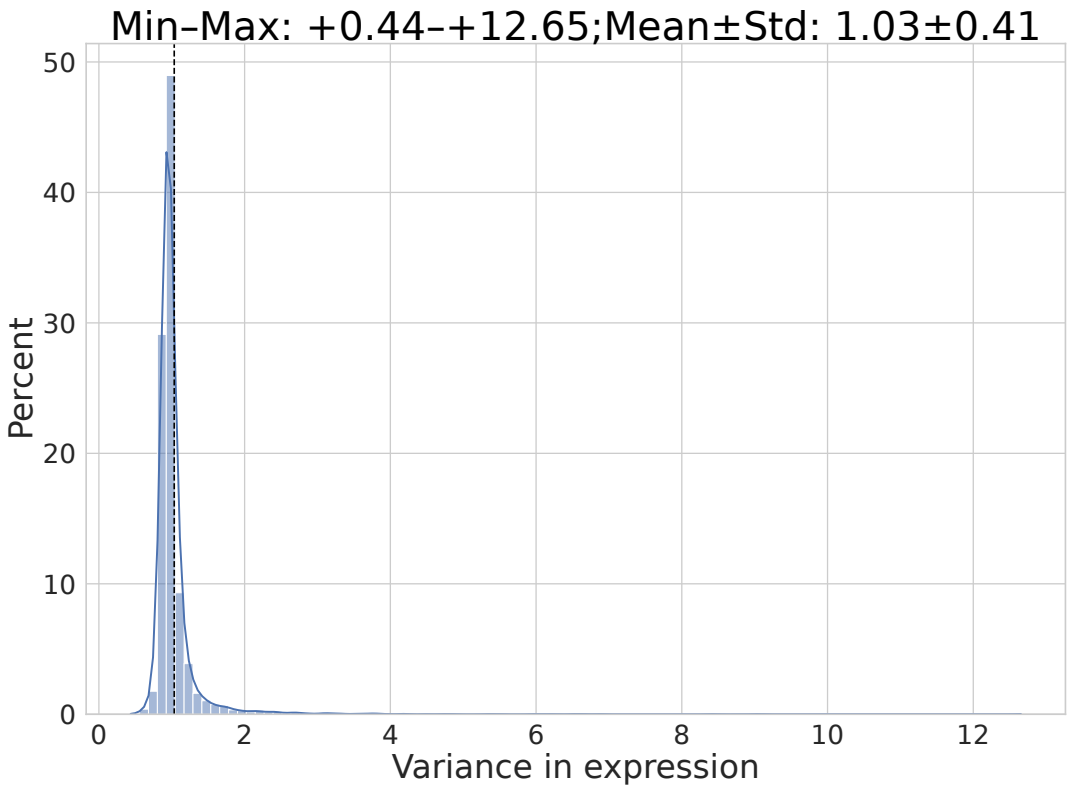


Figure: Histogram

5. Results

5.2. Highly variable genes

5.2.2. Dispersions

Highly variable genes with dispersions I

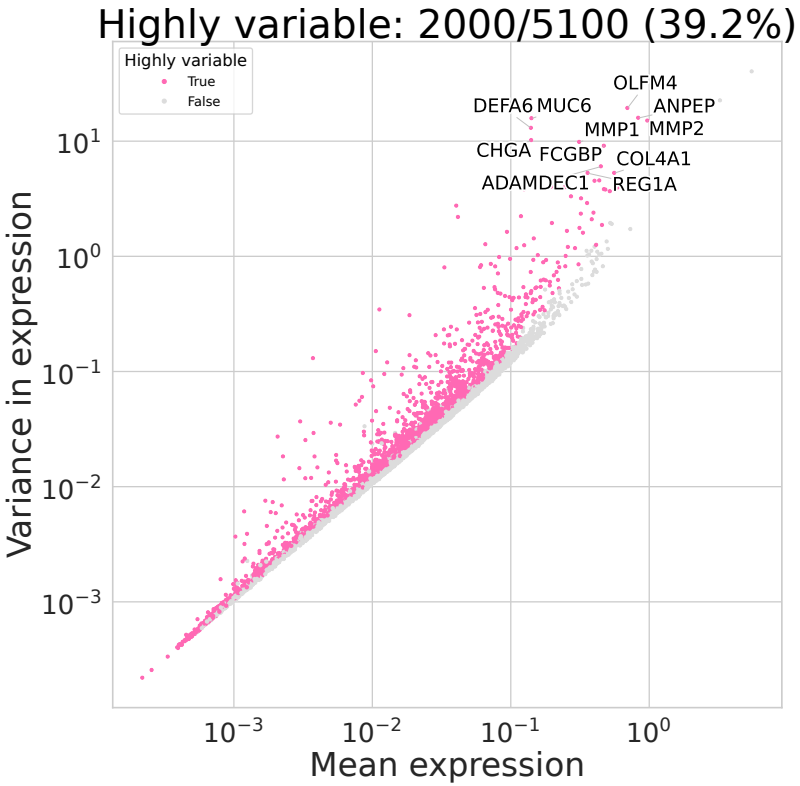


Figure: Volcano plot

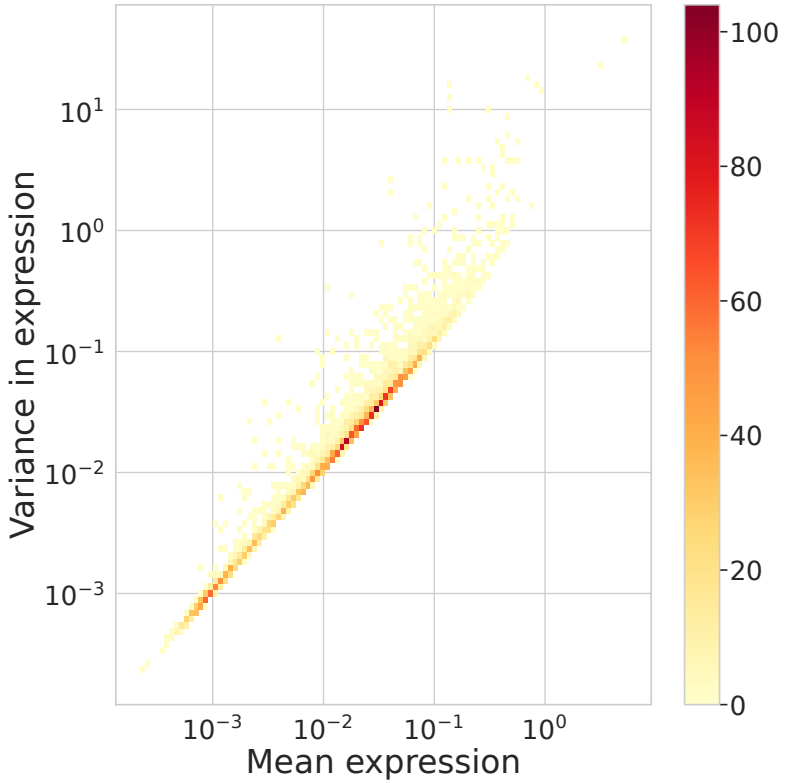


Figure: Distribution

Highly variable genes with dispersions II

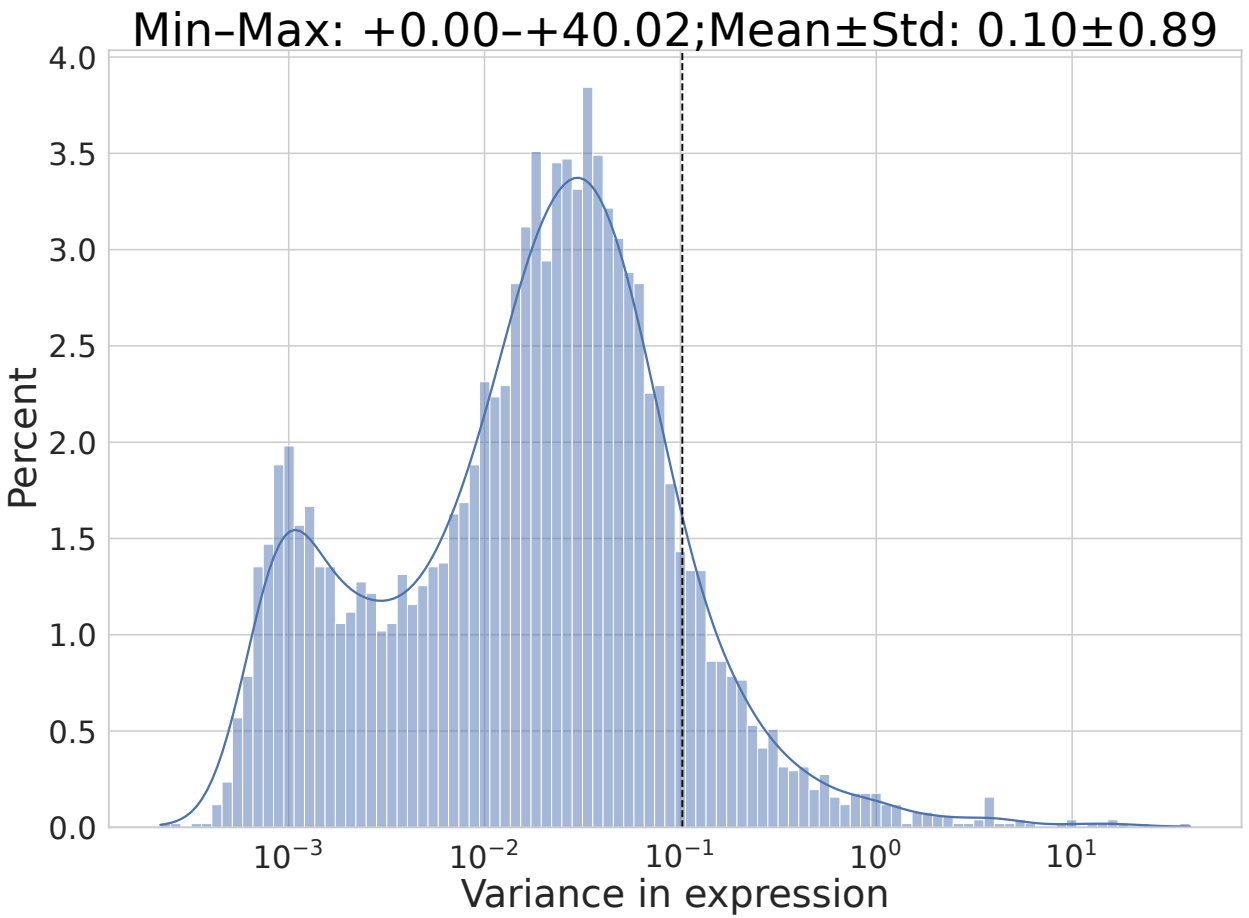


Figure: Histogram

5. Results

5.3. PCA

Explained variance

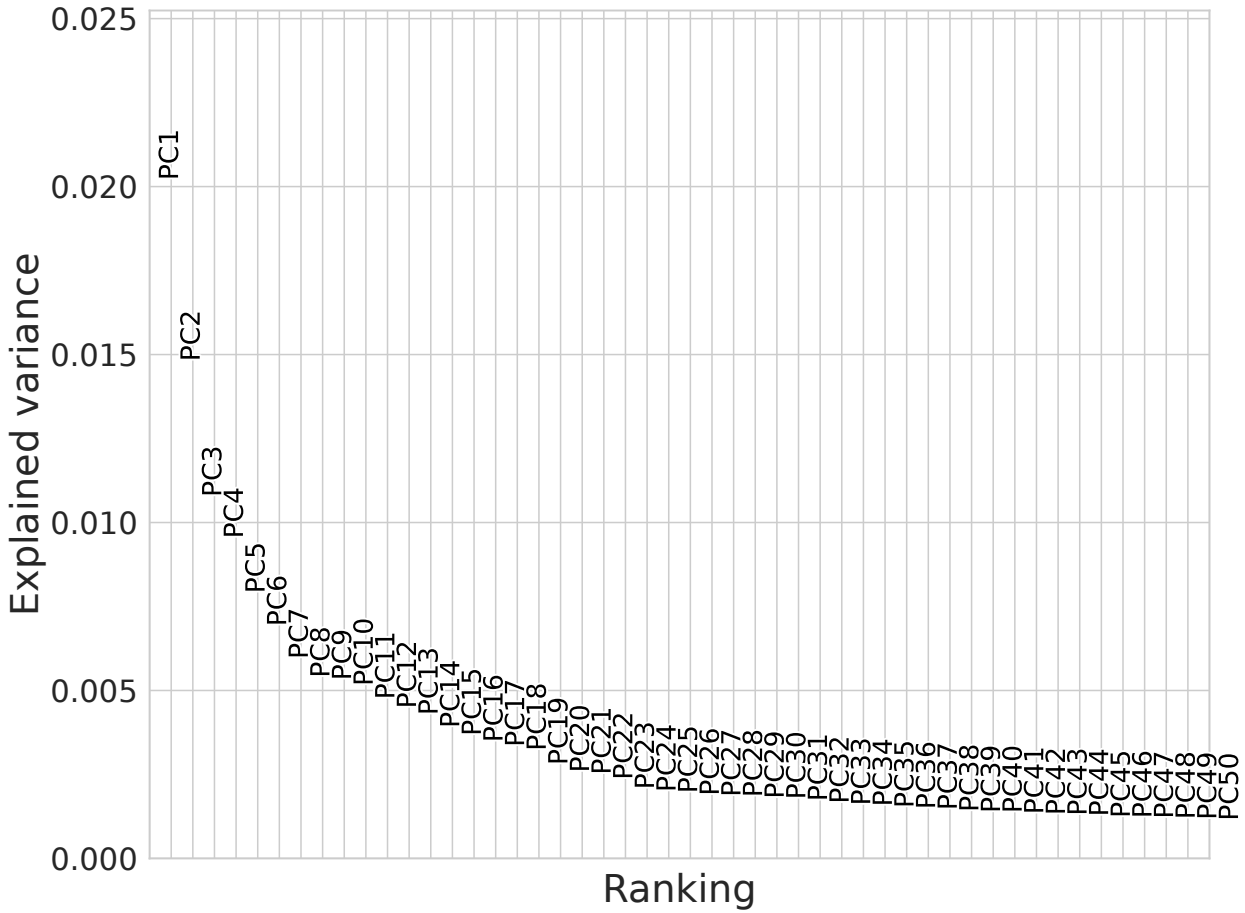


Figure: Explained variances

5. Results

5.4. Clustering

Clustering with UMAP

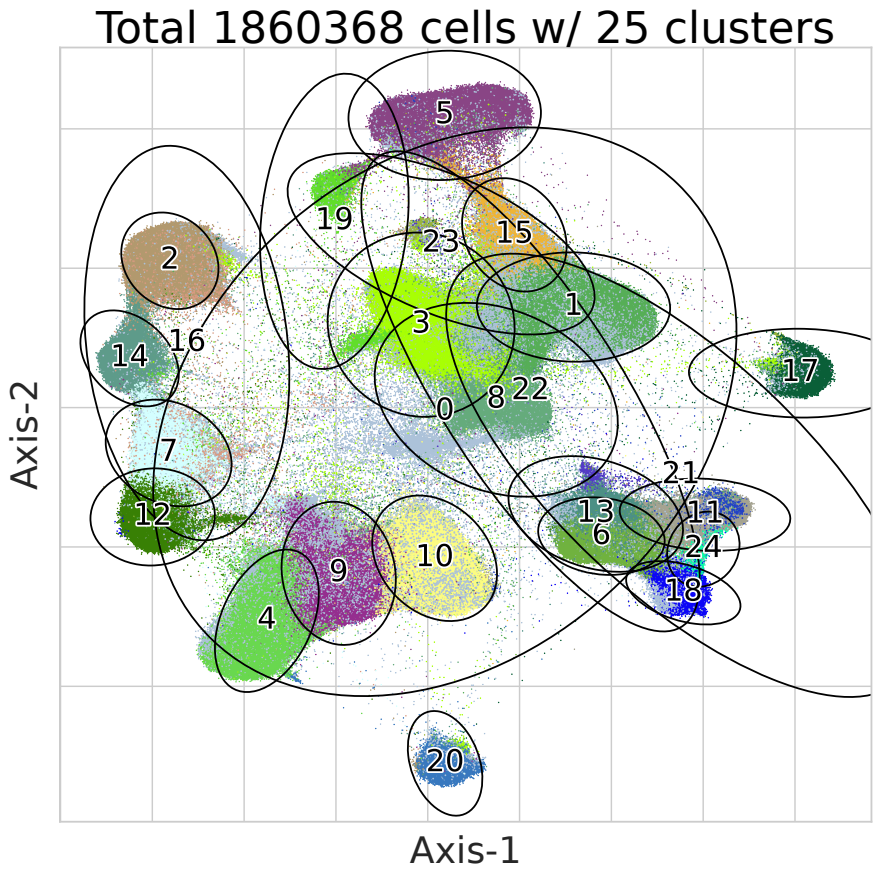


Figure: Clustering with UMAP

Cells per cluster

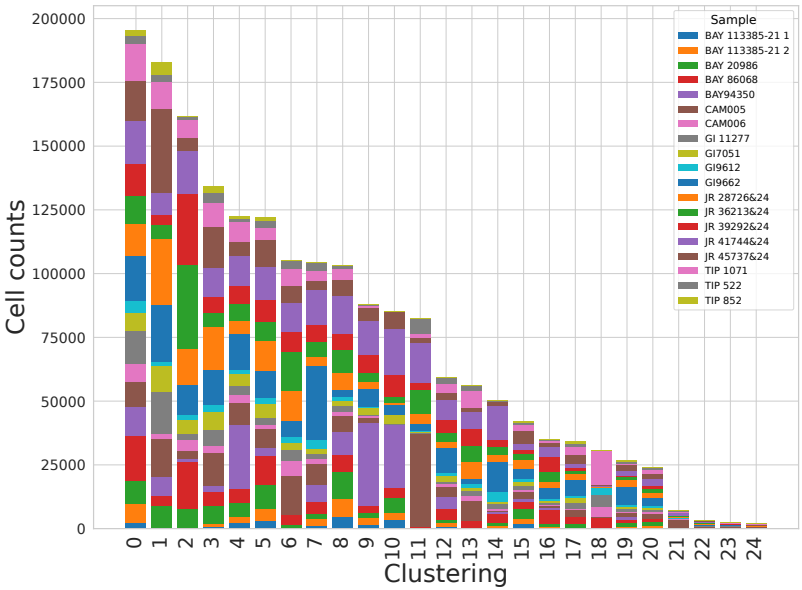


Figure: Absolute count

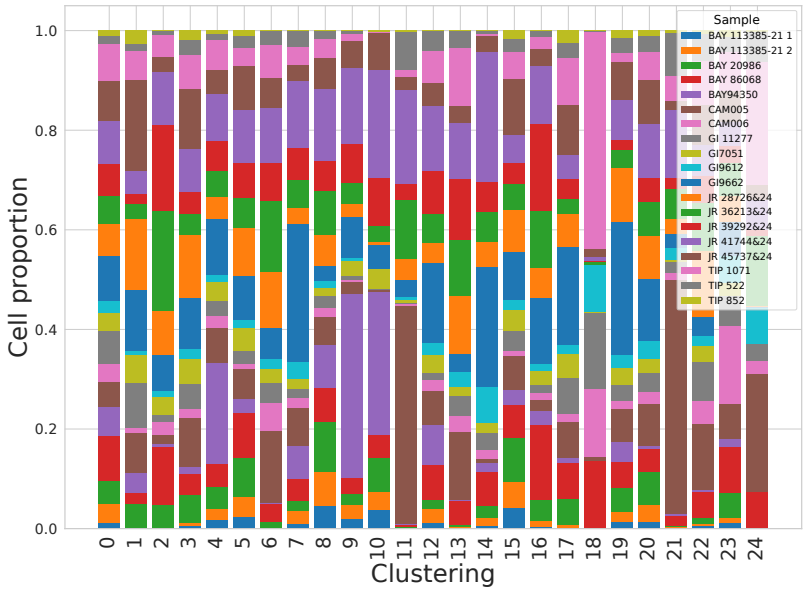


Figure: Relative proportion

Cells per sample

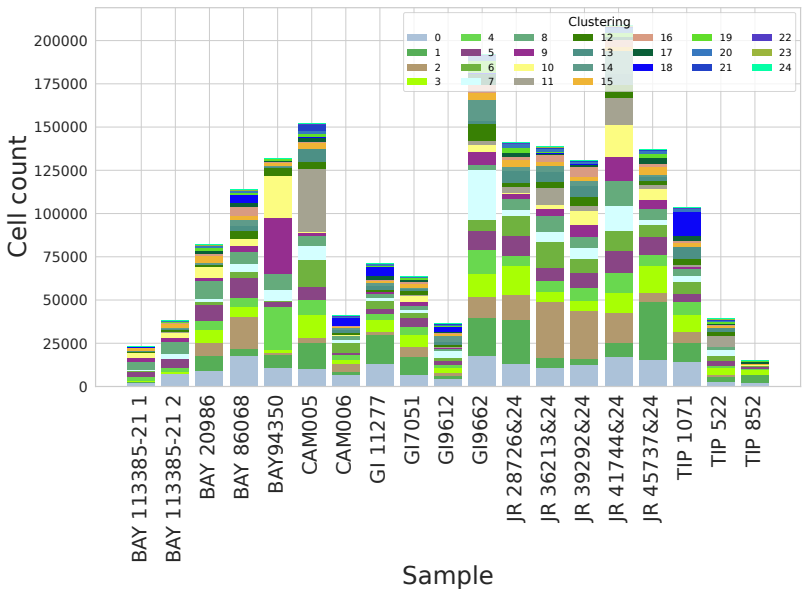


Figure: Absolute count

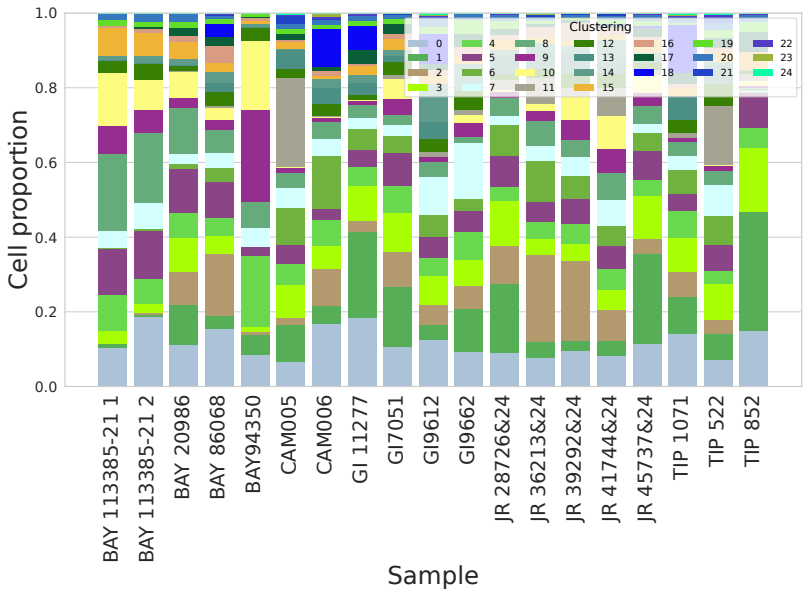


Figure: Relative proportion

5. Results

5.5. Marker genes

Cell type scoring

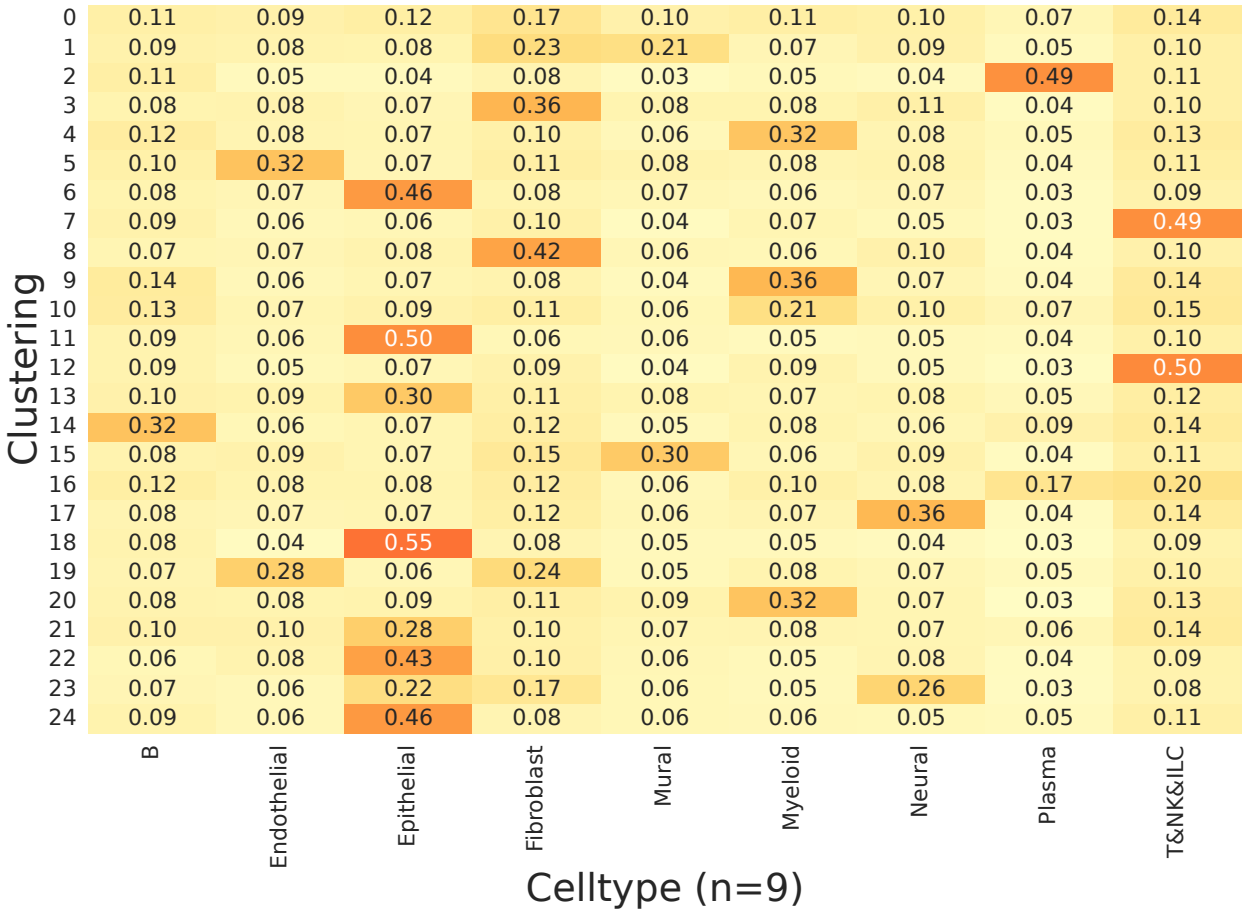


Figure: Cell type scoring

Cell types per cluster I

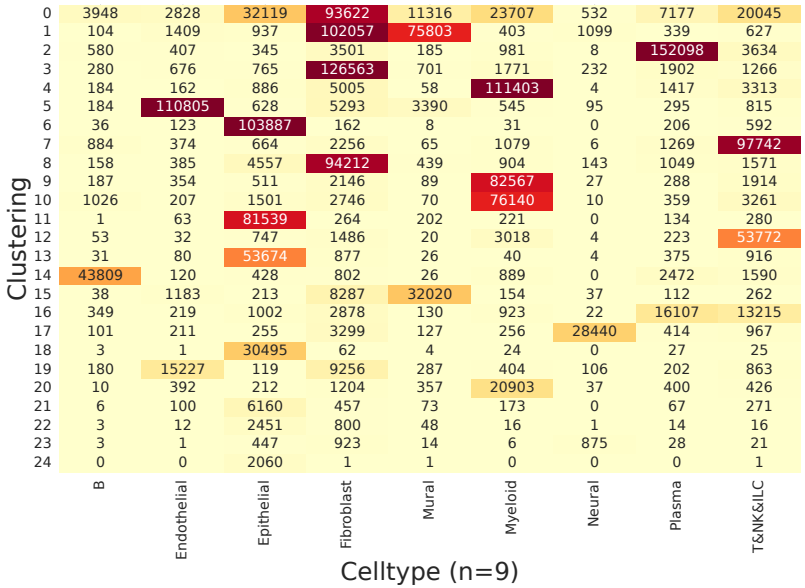


Figure: Absolute count

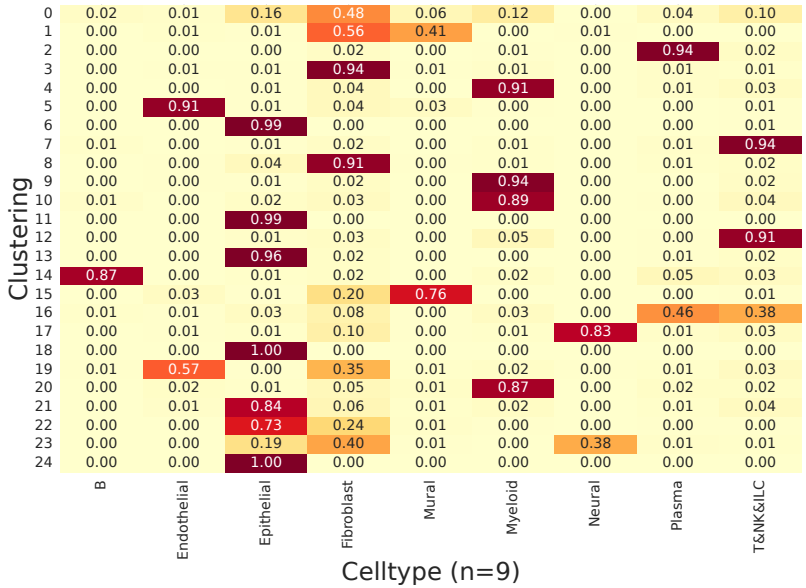


Figure: Relative proportion

Cell types per cluster II

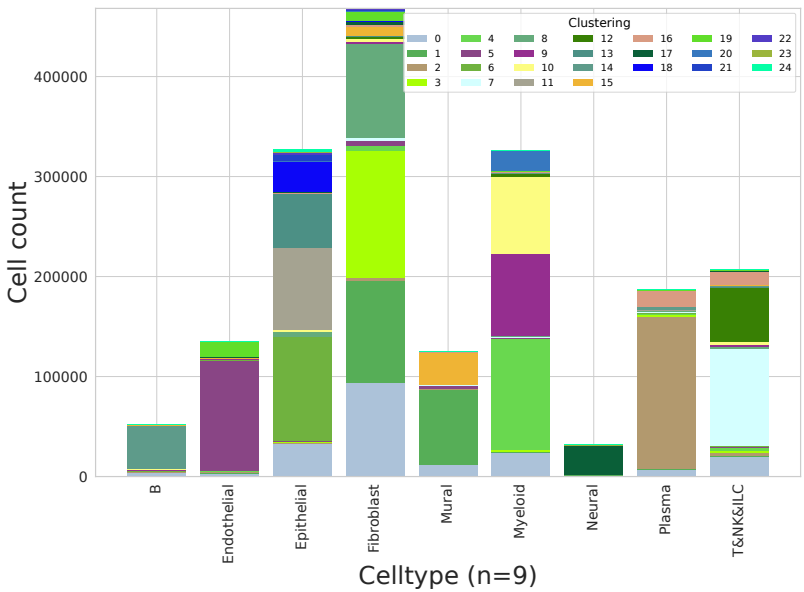


Figure: Absolute count

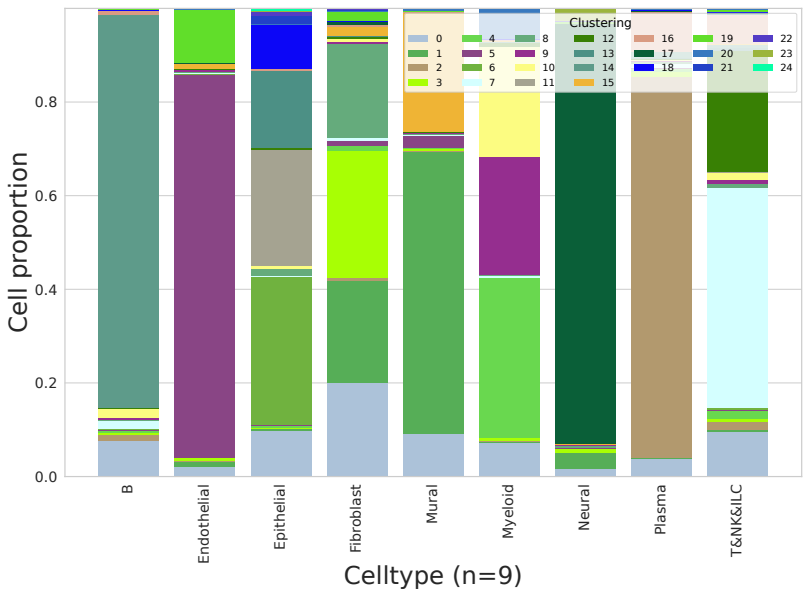


Figure: Relative proportion

Cell types per cluster III

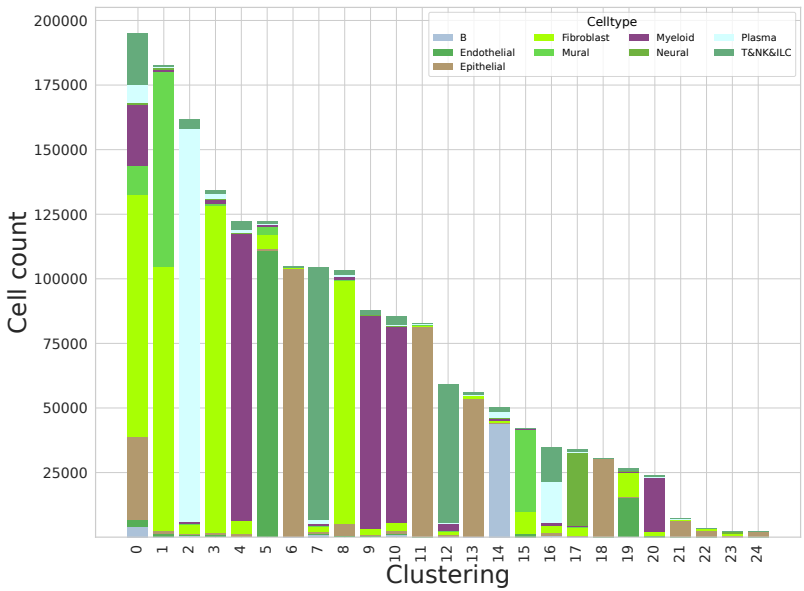


Figure: Absolute count

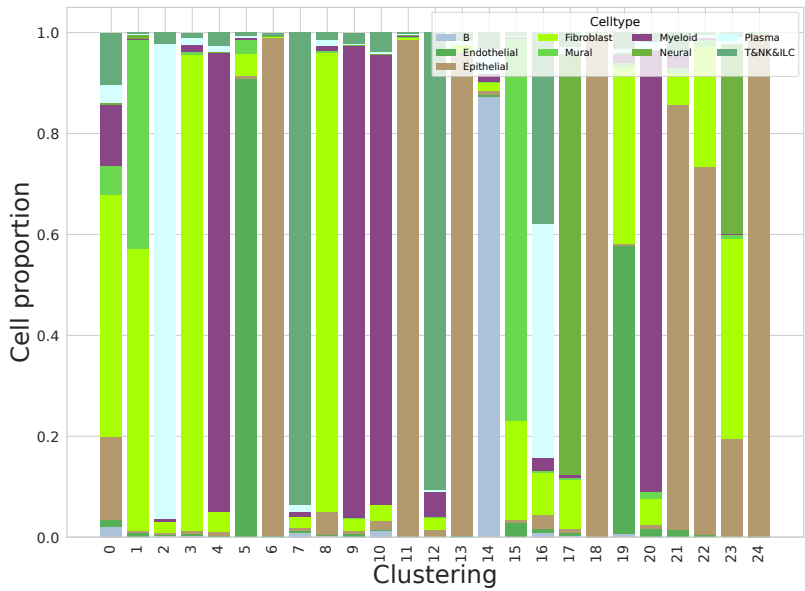
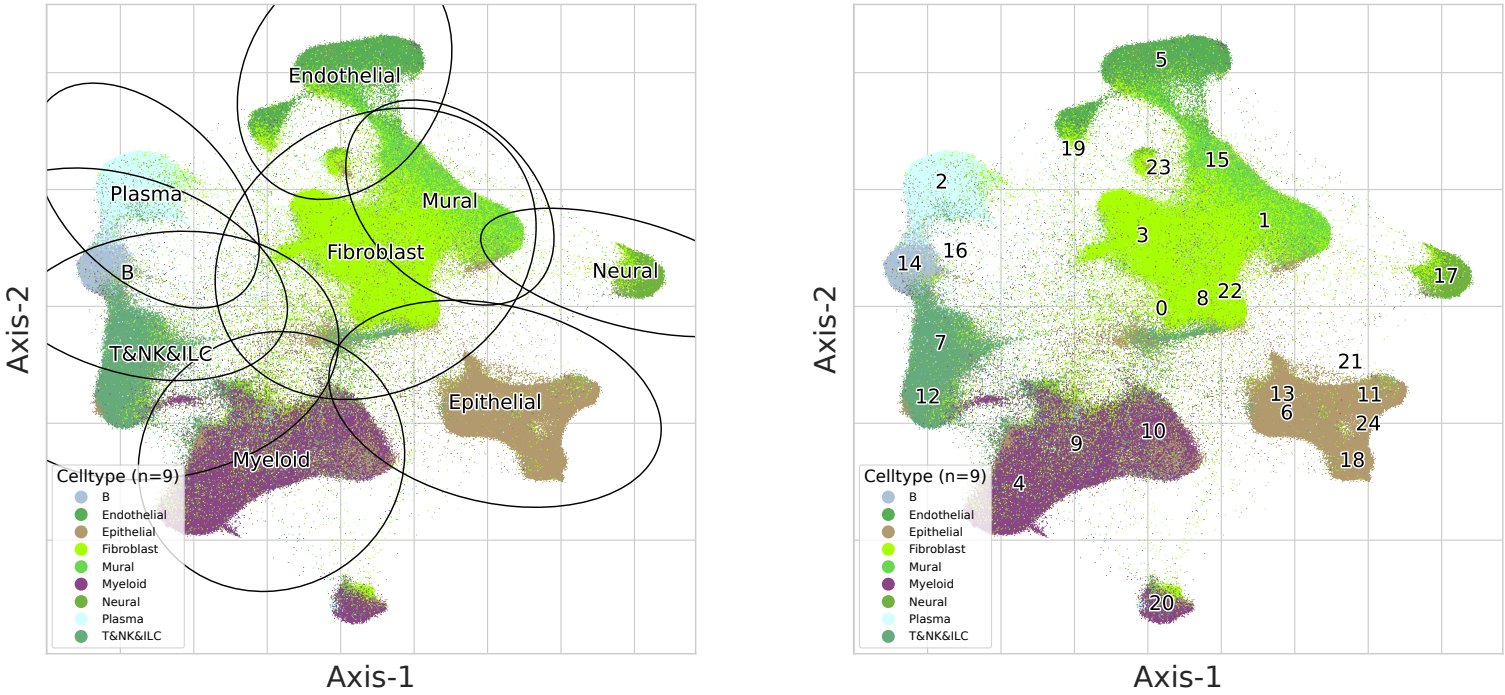


Figure: Relative proportion

Cluster marker genes I



(a) Cell type (b) Clustering

Figure: Cell types with clusters

Cluster marker genes II

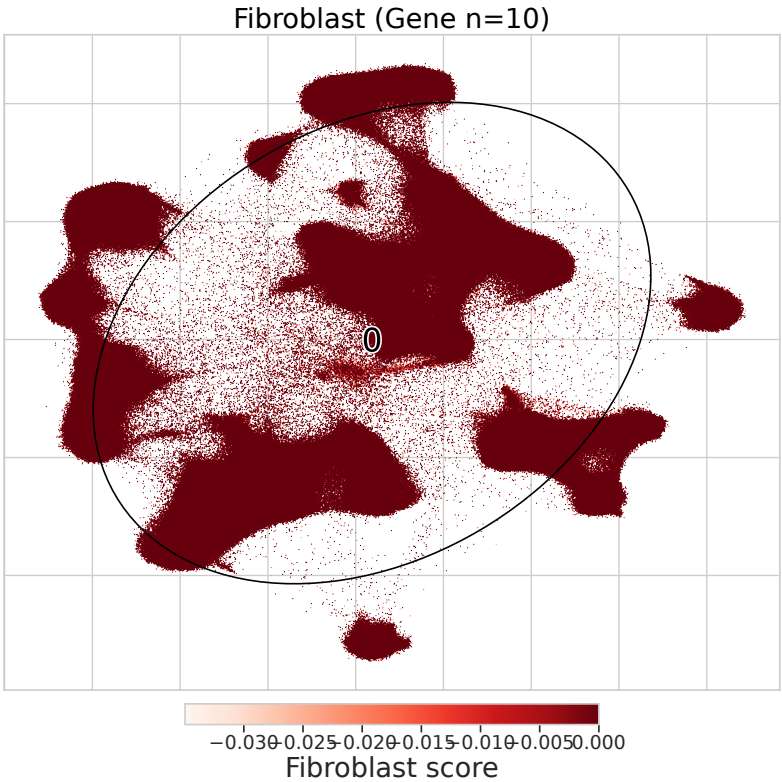


Figure: Cluster #3

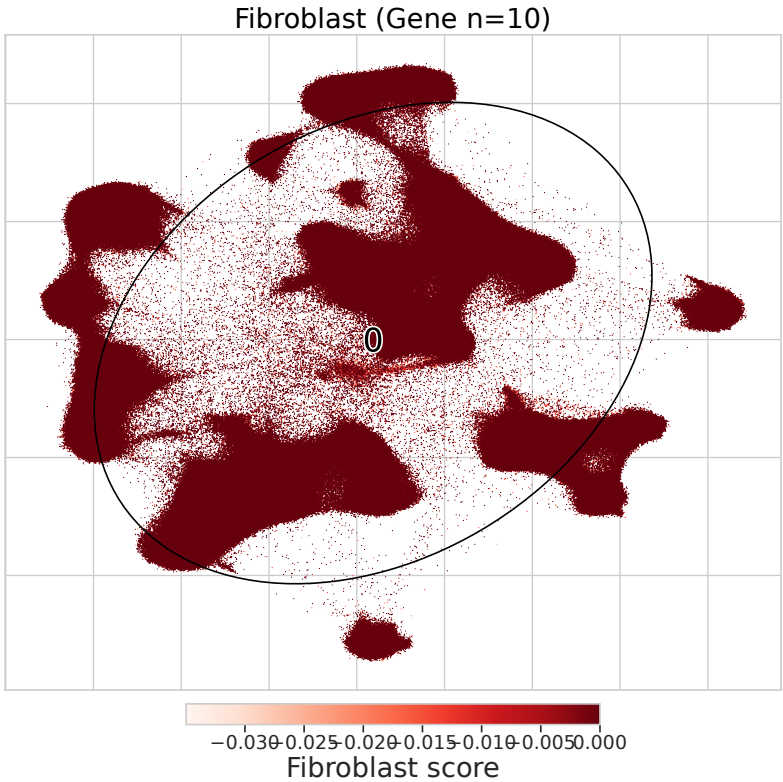
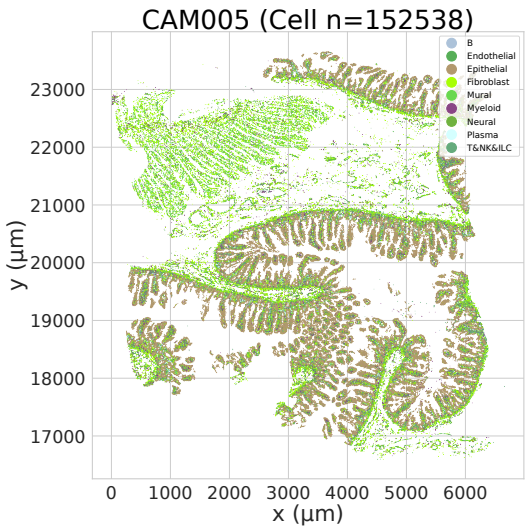


Figure: Cluster #13

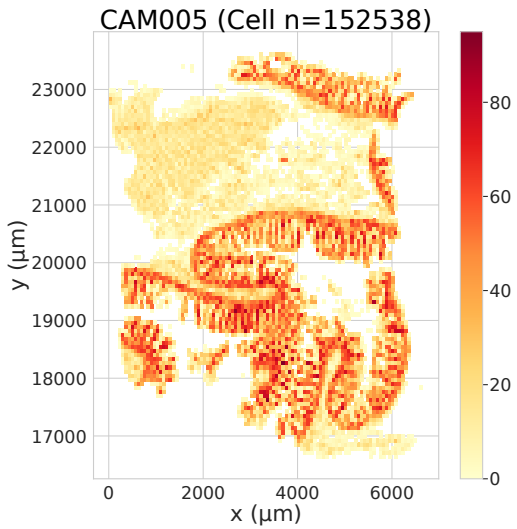
5. Results

5.6. Xenimum coordiantes

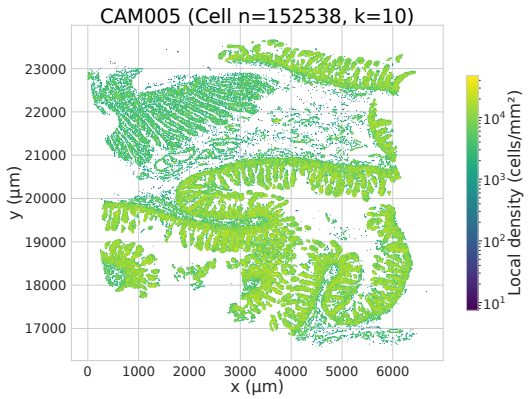
Coordinates of cell types I



(a) Scatter



(c) Hist



(c) Density

Figure: Cell cooridnates in CAM005 sample

Coordinates of cell types II

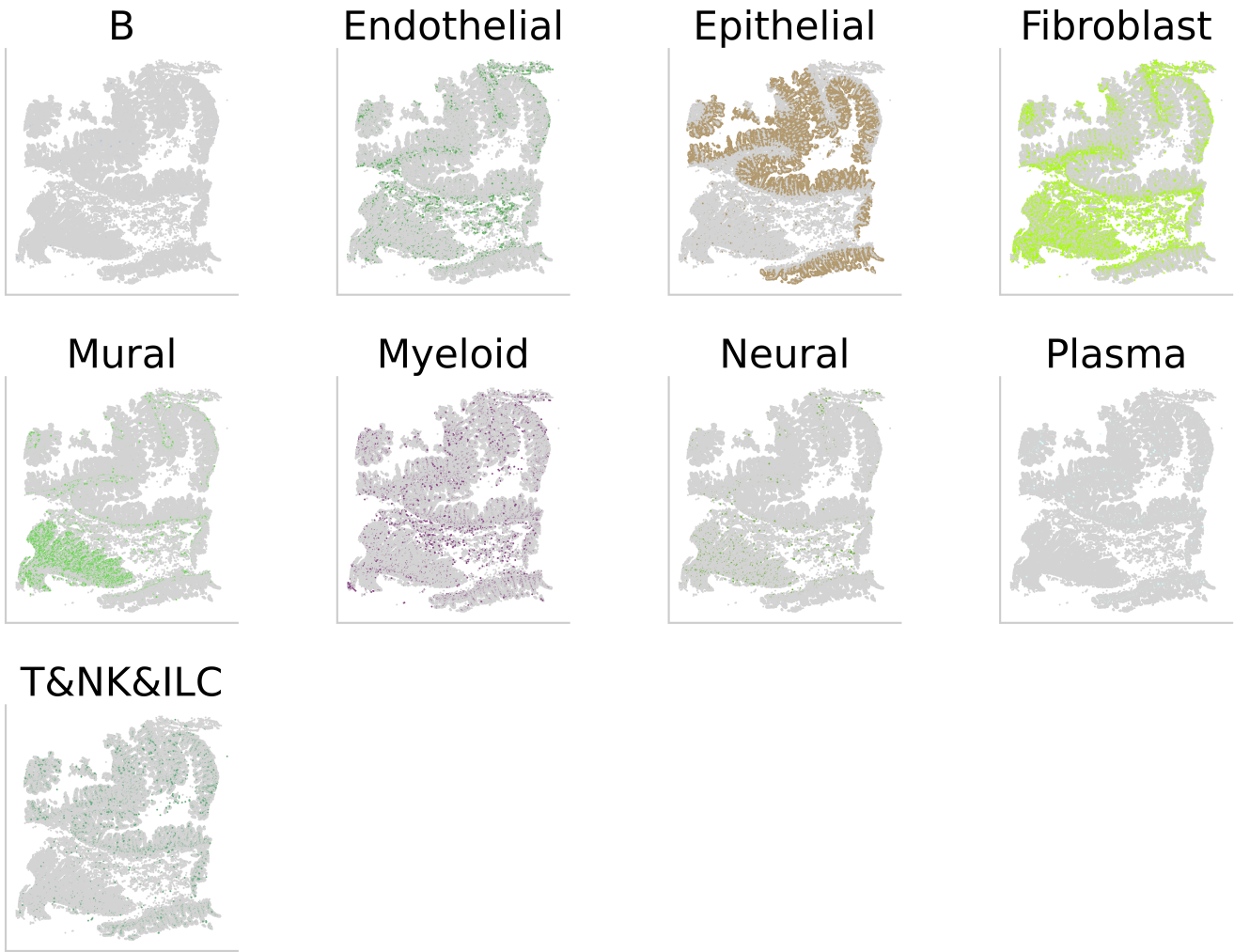


Figure: Coordinates in CAM005 sample

6. References

References I

Amid, E., & Warmuth, M. K. (2019). Trimap: Large-scale dimensionality reduction using triplets. *arXiv preprint arXiv:1910.00204*.

Becht, E., McInnes, L., Healy, J., Dutertre, C.-A., Kwok, I. W., Ng, L. G., ... Newell, E. W. (2019). Dimensionality reduction for visualizing single-cell data using umap. *Nature biotechnology*, *37*(1), 38–44.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment*, *2008*(10), P10008.

Maaten, L. v. d., & Hinton, G. (2008). Visualizing data using t-sne. *Journal of machine learning research*, *9*(Nov), 2579–2605.

McInnes, L., Healy, J., & Melville, J. (2018). Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.

References II

Moon, K. R., Van Dijk, D., Wang, Z., Gigante, S., Burkhardt, D. B., Chen, W. S., ... others (2019). Visualizing structure and transitions in high-dimensional biological data. *Nature biotechnology*, *37*(12), 1482–1492.

Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From louvain to leiden: guaranteeing well-connected communities. *Scientific Reports*, *9*(1), 5233.

Wolf, F. A., Angerer, P., & Theis, F. J. (2018). Scanpy: large-scale single-cell gene expression data analysis. *Genome biology*, *19*(1), 15.